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# Designing a Simulation-based Prototype

## Supporting the implementation of Revenue Models for Deep-Tech Ventures

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Sincerely,

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## Abstract

Deep-tech ventures (DTVs) develop advanced technological solutions that combine technical, physical, and digital elements, creating complex conditions for revenue model implementation. Although there is significant research on revenue model structures, firms in DTV context often lack structural support for comparison and implementation of these revenue models in practice, which creates a design-implementation gap. This study addresses this design-implementation gap by designing a simulation-based prototype to support the comparison and implementation of revenue models within DTVs, integrating both providers perspective and customer insights.

The study adopted design science research approach, using axiomatic design (AD) as the theoretical design framework. The empirical data was collected through consultation with the case company, a Swedish DTV specialized in advanced surface treatment and barrier solutions, complemented by semi-structured interviews with the industrial customers of the case company. The collected data was translated into functional requirements (FRs) and design parameters (DPs), enabling the systematic translation of provider and customer insights into a structured simulation environment for the development of an Excel-based simulation prototype.

The result shows that a modular simulation-based prototype can support the comparison of revenue models regarding profitability, scalability and investment and risk allocation. The study further suggests that revenue models' suitability is dynamic and phase dependent, changing as technology matures and customer trust increases.

The study addresses the design-implementation gap by developing a practical decision-support simulation-based prototype that support the implementation of revenue models in a DTV context.

**Keywords:** *Deep-Tech Ventures, Revenue Models, Product-Service System, Simulation-based prototype, Axiomatic Design*

## Sammanfattning

Deep-tech ventures (DTVs) utvecklar avancerade teknologiska lösningar där fysiska, tekniska och digitala komponenter kombineras, vilket medför komplexitet vid implementering av intäktsmodeller. Trots omfattande forskning kring intäktsmodeller saknar företag i DTV-kontext strukturerad och praktiskt verktyg för att jämföra and implementera dessa modeller, vilket skapar design–implementeringsgap. Denna studie adresserar detta gap genom att utveckla en simulationsbaserad prototyp som möjliggör jämförelse och implementation av olika intäktsmodeller i DTV-kontexter, med utgångspunkt i både leverantörsperspektiv och kundinsikter.

Studien tillämpar en design science research-ansats där axiomatisk design (AD) används som ett teoretiskt designramverk. Studien bygger på empiriska data från konsultationer med fallföretaget som är ett svenskt DTV specialiserad inom avancerad ytbehandling- och barriärlösningar. Denna data kompletterades med semistrukturerade kundintervjuer. Den insamlade data översättes till funktionella krav (FRs) och designparametrar (DPs), vilket möjliggjorde en systematisk översättning av leverantörs- och kundinsikter till en strukturerad simuleringsmiljö för at utveckla en Excel-baserad simuleringsprototyp.

Resultatet visar att en modulär simuleringsbaserad prototyp kan användas för att möjliggöra jämförelse av olika intäktsmodeller utifrån, lönsamhet, skalbarhet och investering och riskfördelning. Studien föreslår att hur lämplig en intäktsmodell är varierar över tid och är fasberoende. Det innebär att valet av intäktsmodell är kopplad till teknologins mognadsgrad och förändras när teknologin mognar och kundförtroende ökar.

Studien bidrar till att minska design-implementationsgapet genom att utveckla ett simuleringsbaserat beslutstöd prototyp som möjliggör jämförelse och implementering av intäktsmodeller i DTV-kontext.



**Nyckelord:** *Deep-Tech Venture, Intäktsmodeller, Produkt-tjänstesystem, Simuleringsbaserad prototyp, Axiomatisk Design*

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## Abbreviations

| | |
|---|---|
| DTV | Deep-Tech Venture |
| PSS | Product-Service System |
| AD | Axiomatic Design |
| FR | Functional Requirement |
| DP | Design Parameter |

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# 1 Introduction

*This section contains a background, problem formulation, purpose and research question, and case description.*

## 1.1 Background

Advanced groundbreaking technologies are increasingly essential for societal and industrial development, as they support sustainability competitiveness and the ability to address complex global challenges (Vinnova, 2024; Raff et al., 2024). Many of these advanced technologies are developed by Deep-tech ventures (DTVs) which are research-intensive startups that develop advanced technological solutions based on advanced scientific research (Raff et al., 2024; Romme et al., 2023). DTVs are becoming increasingly vital to the sustainability transition by developing technologies that optimize resource consumption, reduce industrial emission and support more sustainable industrial system (Vinnova, 2024). These ventures support industrial transformation through new technological applications and business opportunities across multiple industries (Grande et al., 2025). Furthermore, these advanced offerings often combine technical, physical, and digital elements, creating opportunities for several revenue model structures such as asset sales, leasing, licensing and subscription-based models (Tukker, 2004; Osterwalder & Pigneur, 2010).

Despite their technological capabilities, advanced technological offerings developed by DTVs may face challenges related to implementation and sustained market acceptance (Raff et al., 2024). Because these technologies are often novel and lack clear market benchmarks, firms may experience challenges related to implementation, validation and long-term market adoption (Romme et al., 2023; Raff et al., 2024). Furthermore, customers may experience difficulties assessing the long-term value, operational implications and financial investments requirements related to advanced technological offerings (Bonnemeier et al., 2010; Yang & Evans, 2019). Consequently, firms need to implement revenue models capable of structuring ownership, operational responsibility, and risk sharing-sharing between providers and customers (Yang & Evans, 2019; Teece, 2010). Dmitriev et al. (2014) further argue that the success of advanced technological offering depends not only on technology itself, but also on how firms' structure and monetize the value through suitable revenue models. Consequently, revenue models become essential mechanism for balancing value capture, operational responsibility and long-term implementation between provider and customer (Bonnemeier et al., 2010; Yang & Evans, 2019).

How firms capture value from their advanced technological offerings is largely determined by the structure of revenue model (Teece, 2010; Remeňová et al., 2020). Different revenue models differ in how ownership, responsibility as well as value capture is structured

between provider and customer. However, identifying and implementing suitable revenue models for advanced technological offerings remains challenging, since firms need to balance value capture and profitability with customers' demands regarding, investments, operational adaptability and risk reduction (Linde et al., 2023; Bonnemeier et al., 2010). Consequently, firms require more structured approaches for assessing and comparing how different revenue models affect profitability, operational responsibility and implementation conditions between providers and customers.

### 1.2 Problem Formulation

Existing research discusses several revenue model options for advanced technological offerings, including asset sales, leasing, subscriptions as well as result-oriented models, which differ in distributions of ownership, risk and value between providers and customers (Tukker, 2004; Yang & Evans, 2019; Linde et al., 2023). Furthermore, research suggests that service-oriented and recurring revenue models can strengthen customer relationships while creating long-term revenue streams (Kowalkowski & Ulaga, 2017; Tukker, 2015). Meanwhile, these models frequently require providers to assume higher financial and operational responsibility, thereby increasing implementation complexity (Mont, 2002; Yang & Evans, 2019).

However, despite extensive literature discussing revenue models and value capture mechanisms, implementing suitable revenue models remains challenging in practice. These implementation challenges are particularly critical in DTV contexts characterized by advance technological integration, long validation cycles and lack of clear market benchmarks (Raff et al., 2024; Romme et al., 2023). This challenges firms' ability to assess how revenue models should structure ownership, operational responsibility and investment during different phases of implementation. Consequently, firms in DTV context may face challenges identifying revenue model structures that are operationally feasible and financially suitable under evolving implementation conditions (Raff et al., 2024; Romme et al., 2023). Traditional decision-making approached may therefore become insufficient, as implementation conditions, customer requirements and operational responsibility evolve over time (Bonnemeier et al., 2010; Yang & Evans, 2019).

Revenue models that are theoretically attractive may prove challenging to implement in practice, while revenue models that are strategically attractive from a customer's perspective are not necessarily aligned with providers requirements regarding profitability, operational responsibility and scalability.

Different revenue models distribute ownership, operational responsibility, investment requirements and risk between providers and customers, thereby complicating implementation decisions (Yang & Evans, 2019; Bonnemeier et al., 2010). Therefore, firms may implement revenue models that fail to balance customer expectations, willingness to

pay, operational capabilities, and long-term scalability. Consequently, the challenge is not primarily about identifying new revenue models, but rather supporting decision-making related to comparison and practical implementation of existing revenue models.

This creates a design implementation gap, where existing literature identifies different revenue models and value capture mechanisms, but offer limited support regarding how revenue models should be assessed, compared and implemented in practice (Chesbrough, 2010). Existing literature primarily focuses on revenue models and value capture but offers limited guidance on practical implementation comparison and provider and customer alignment within DTV (Teece, 2010; Tukker, 2004). Although pervious research discusses revenue model structures (Linde et al., 2023) and simulation-based business model innovation (Ksouri-Gerwien & Poeppelbuss, 2024), there is still limited research on how firms can practically compare and assess revenue model alternatives.

These challenges suggest a potential need for structured simulation-based decision support tool that can support the comparison of revenue models. By visualizing how different revenue models affect cost, revenue, operational responsibilities and long-term value creation, such tool may support firms in understanding the implication of different alternatives. Furthermore, simulation-based comparison may support alignment between providers strategic goals with customer requirements regarding investment, operational adaptability and risk sharing (Linde et al., 2023). Despite this potential, current research offers limited practical support explaining how such tools can be designed and utilized to support revenue model comparison and implementation.

Without structured simulation tools, managers and firms risk relying on assumptions or fixed static decision processes when choosing revenue models. This increases the risk of selecting revenue models that financially impractical, operationally unsuitable or misaligned with customer requirements (Chesbrough, 2010; Linde et al., 2023). Consequently, there is a growing need for tools that supports managers and firms in comparing how various revenue models impact profitability, scalability and risk allocation.

### 1.3 Purpose and Research Question

The purpose of the study is to design a simulation-based prototype for supporting comparison and implementation of revenue models within DTV by integrating providers perspective and customer insights. The prototype aims to support managerial decision-making by enabling comparison of revenue models in the terms of profitability, scalability and risk-sharing, thereby contributing to narrowing design implementation gap.

To address the identified problem, the following research question is formulated:

RQ: *How can a simulation-based prototype be designed to support the comparison of revenue models in DTVs from the provider perspective and insights from their customers?*

This study focuses on designing simulation-based prototype that supports the implementation of revenue models through comparison within DTVs. This study does not seek to develop new revenue models theories or provide a broader explanation of organizational strategy or customer behaviour. Instead, it is limited to implementation of existing revenue models, taking to account both providers perspective and customers insights.

### 1.4 Case Description

The study is based on the Swedish Deep-tech venture, a firm developing advanced surface treatment and barrier solutions within the advanced material field. The firm's technology is grounded in scientific research and is developed to protect industrial components from corrosion, wear and hydrogen-related deterioration. By applying specialized treatment processes as well as patented material technology, the firm's aims to extend component lifetime, minimize maintenance needs and improve resource efficiency in industrial operations. The company mainly operated within industrial manufacturing sectors, were customer places strong focus and emphasis on long-term product performance and operational reliability and sustainability. Its offerings integrate advanced material technology, industrial treatment processes, service-oriented activities and products for the treatment processes, creating possibilities for several revenue model structures.

This case is theoretically relevant as it mirrors several characteristics associated with DTVs and advanced technological offerings. First, the company develops a technologically advanced and research-intensive solution based on patented material technologies, which aligns with existing literature describing DTVs as firms operating in environments combining advanced science, industrial applications and technological innovation (Raff et al., 2024; Romme et al., 2023). Second, the technology can be monetized through several revenue model alternatives, including asset sales, licensing, subscription services, and recurring revenues tied to consumable products. These alternatives differ in how ownership, operational responsibility, investments, and value capture are distributed between provider and customer, thereby creating implementation challenges regarding profitability, scalability and customer adaptation. Consequently, the case offers a relevant context for studying how simulation-based decision-support can support firms in comparing and implementing revenue models for advanced technological offerings within DTV environments.

Furthermore, the case demonstrates several implementation challenges discussed in existing DTV literature. Compared to more established industrial firms, this company operates in an early-stage industrial environment characterized by lack of clear market benchmarks for performance and implementation of different revenue models and evolving customer requirements. Furthermore, the firm's advanced technology has

potential applications across several industrial sectors, where customer requirements, operational environment and investment capabilities differ substantially. This creates challenges in understanding how different revenue models influence profitability, scalability, operational responsibilities, and customer requirements. Consequently, selecting and implementing suitable revenue models becomes a complex decision-making process shaped by various interrelated factors related to the firm and customers.

This study is therefore valuable for the firm as it contributes to a deeper understanding of how various revenue models can be compared and implemented within DTV environments. Specifically, the study focuses on developing a simulation-based prototype that can visualize and compare how different revenue models influence profitability, operational responsibility investments and risks are allocated between provider and customer. By simulating different revenue model alternatives, the company may improve its ability to assess how ownership structures, operational responsibilities, revenues, investments, and risks are allocated between the company and customer. Building upon this, the study focuses on the firm as the case company where different revenue model alternatives can be compared and implemented through the development of a simulation-based prototype.

# 2 Theoretical Framework

*This section outlines the theoretical foundation for the study, bridging the characteristics of deep-tech ventures with product-service systems and revenue models. It outlines how these elements introduce a design-implementation gap and presents axiomatic design as the systematic framework used to address this complexity.*

## 2.1 Deep-Tech Ventures

Deep-tech ventures (DTVs) are new startups grounded in scientific breakthroughs and advanced technologies, often emerging from academic or institutional environments such as universities or specialized research institutes (Raff et al., 2024). These ventures focus on developing advanced technological solutions for applications including energy systems, new materials and high precision engineering systems. DTVs advanced technological offering is often exposed to technological complexity as well as long development process (Raff et al., 2024, Romme et al., 2023).

Deep-tech startups are another term that may be confused with DTVs. Although both terms refer to new established firms characterized by long development cycles and high risk, a distinct difference exists between them. While deep-tech startups refer to research-intensive firms in an early stage of development (Schuh et al., 2022), DTVs emphasize the broader venture, including market implementation, scaling challenges and the external capital required to reach the market (Raff et al., 2024; Romme et al., 2023). This distinguishes them not only by their technological foundation but also in their strategies for value creation and commercialization.

Because advanced technological offered by DTV are innovative and technically complex, firms and customers may face challenges in assessing operational implications, long-term value and implementation requirements (Raff et al., 2024; Romme et al., 2023). Therefore, firms may need more dynamic and flexible revenue models to enable implementation and long-term value capture.

Furthermore, several contextual and organizational factors affect how DTVs commercialize their technologies as well as implement revenue models. Pervious research identifies factors such as technological maturity, financial limitations, ecosystem dependencies, institutional conditions, competitive pressure and market structure that shape commercialization strategies (Raff et al., 2024; Romme et al., 2023; Chesbrough, 2010). For example, DTVs often require support from external actors, investors and industrial partners because of limited internal resources and complex development processes. At the same time, extended validation cycles and uncertain market adoption can complicate the assessment of how value should be delivered and captured over time. Although, this study does not empirically explore these factors, they offer a broader understanding of why revenue model implementation in DTV is complex.

Another defining characteristic of DTVs is their strong reliance on propriety and patent protected technologies. Because these technologies are research intensive, firms deliberately obtain intellectual property rights to protect and capture value from their technological advancement (Raff et al., 2024). Patents provide exclusive rights to technology, allowing the focal firm to either exploit it internally or selectively license access to external parties under defined conditions (Woo & Park, 2025). This exclusivity is critical, as patents reflect the outcome of extensive research efforts that drive technological advancements and economic returns (Teece, 1986). Consequently, advanced technologies offerings by DTVs may require revenue models that enable implementation, long-term value capture and satisfy both provider and customer.

Given that advanced technological offerings in DTVs often involve the integration of products, services, technical support and long-term operational responsibilities, firms need to decide how value should be created deliver and monetized. Therefore, product-service system (PSS) becomes particularly relevant in DTV, as they provide a framework for understanding how integrated offerings combine products and services while defining ownership, operational responsibilities as well as value capture between provider and customer (Tukker, 2004; Mont, 2002).

### 2.2 Offerings in Deep-Tech Ventures

As advance technological offerings are ready to become implemented, firms need to determine how value should be structured, delivered and monetized in practice. In this context, PSS function as "mechanism" for structuring, delivering and communicating value through the integration of product and services into an integrated solution (Tukker, 2004; Mont, 2002). This perspective is highly relevant for advanced technological offerings in DTVs, as these offerings often integrate physical products together with service activities and technical support rather than functioning as isolated products. This view is further supported by existing literature which position PSS as a strategic mechanism for structuring value proposition and value communication (Baines et al., 2007; Kowalkowski & Ulaga, 2017). Consequently, PSS are highly relevant for structuring how revenue models are implemented where it satisfies both providers goals and customers' needs.

According to Tukker (2004), offerings are categorized into three architectures of PSS: product-oriented, use-oriented and result-oriented – which primarily differ in how value is created, who owns the product, how risk is allocated and how value is delivered and monetized between customer and provider. Therefore, PSS architectures and revenue model implementation are closely connected, as different offering structures shape different requirements for profitability, scalability and operational responsibility, making revenue model implementation highly context dependent (Yang & Evans, 2019; Barquet et al., 2011).

Moreover, product-oriented PSS implies selling a physical product and complements it with various services such as maintenance, support, or consulting. The customer gets the ownership of the product, and value creation remains largely tied to the tangible artifact. Consequently, services function to enhance the product's lifespan, functionality, and overall customer value (Tukker, 2004; Barquet et al., 2011). An advantage of this offering is that it does not require significant capital commitment and involves relatively low risk for the firm. However, it often limits scalability and differentiation, since the value proposition remains strongly linked to the product itself (Tukker, 2004). Furthermore, with ownership transferred to the customer, most financial and operational risks become customers responsibility, which lowers the risk for the provider (Tukker, 2004; Barquet et al., 2011; Yang & Evans, 2019).

However, this offering introduces several limitations. First, it limits the generation of recurring revenues, as the value is captured mainly at the point of transaction (Barquet et al., 2011). Second, it limits the interaction with the customer which reduces the opportunities for feedback and learning, which are crucial in deep-tech environments (Raff et al., 2024). Finally, this offering offers limited flexibility to evolving customer needs as well as technological advancements, which can affect long-term competitiveness (Tukker, 2004).

In contrast to product-oriented PSS, use-oriented imply that the provider retains ownership of the product and instead sells access to or usage of it. Here, value shifts from ownership to utilization, creating incentives for the firm to optimize product usage, lifespan, and efficiency (Tukker, 2004; Barquet et al., 2011). One of the benefits of this offering lies in its ability to reduce adoption barrier for customers by removing the necessity of substantial initial investments Mont, 2002). Romme et al. (2023) adds that this offering pave the way for continues interaction between the provider and customer which enables firms to gather feedback data and improve learning and facilitate iterative development and adapt to changing customer needs.

However, this offering presents substantial challenges. The necessity for the provider to retain ownership leads to greater capital investments and financial risk, posing challenges for DTVs with limited resources (Tukker, 2004). Mont (2002) adds that operational complexity rises, as firms must coordinate maintenance, logistics as well as continuous service delivery. Barquet et al. (2011) further adds that there is a risk of revenue streams becoming volatile due to variations in usage, adding to financial complexity.

Moreover, result-oriented PSS, represents the most advanced form compared to the two abovementioned. In this architecture, customers do not pay for the product or its use, but rather for the outcome or function delivered (Tukker, 2004). The product itself becomes secondary, while the primary focus is on the performance achieved, such as energy efficiency, productivity improvements, or specific outcomes (Tukker, 2004; Barquet et al.,

2011). This approach entails a high level of abstraction and typically requires the provider to assume greater responsibility for the performance of the entire system. Consequently, result-oriented services are particularly relevant, as customers pay for the impact generated by the technology rather than the technology itself. A positive aspect of this offering is the close alignment of incentives between provider and customer, as both derive value from better performance outcomes (Tukker, 2004). This can build trust and support adoption, particularly when improvements are quantified and explicitly connected to value creation. Additionally, this offering allows firms to capture increased value by tying revenues directly to delivered results instead of inputs or levels of usage (Barquet et al., 2011).

However, this imposes a high risk for providers. By taking on outcome responsibility, they are exposed to uncertainties related to technology performance, operational conditions as well as external factors outside its control (Tukker, 2004). Furthermore, result-oriented offering relies on advanced capabilities in monitoring data analytics, and system integration, which are often not available in early-stage venture (Mont, 2002). The author adds that pricing this offering can be complex, as it requires reliable evaluation and attribution of value, which is often challenging.

Consequently, different PSS structure establishes distinct conditions for implementing revenue models by shaping ownership, profitability, operational responsibility and value capture between customer and providers (Tukker, 2004; Yang & Evans, 2019). This creates a need for tools and mechanisms for comparing different revenue models with the integration of providers goals and customer's needs.

### 2.3 Revenue Models

Various terms in the literature are used to describe how a firm capture value – including business, revenue or price models and how firms earn money (Teece, 2010; Osterwalder et al., 2005; Zott & Amit, 2010; Applegate, 2001; Magretta, 2002). This study uses the term revenue model, referring to how a firm generate income from their products or services and through different monetization structures (Teece, 2010). In line with Tabares et al. (2023), it represents the mechanism through which value is monetized, linking directly to the value capture dimension. Therefore, revenue models play a key role by enabling firms to align their value propositions with sustainable and scalable revenue streams (Barquet et al., 2011; Zott & Amit, 2010).

Revenue streams, in turn, refer to the specific ways in which a firm generates revenue from its value proposition (Osterwalder & Pigneur, 2010), and can be understood as the operational building blocks of the revenue model (Remeňová et al., 2020). According to Osterwalder & Pigneur (2010), revenue streams can be generated through different ways; asset sale, usage fee, subscription fee, lending/leasing/renting, licensing, brokerage fees and advertising – which can be classified as transaction or recurring revenues. Scholars argue

that the more service-driven the approach, with recurring revenues embedded in the revenue models, the better (Tukker, 2004; Tukker, 2015), as such approaches are associated with higher customer satisfaction and profitability (Kowalkowski & Ulaga, 2017). However, it requires firms to balance between customer affordability, perceived value and risk exposure (Mont, 2002; Zott & Amit, 2010). Simultaneously, these approaches also place greater responsibility on the provider, creating higher risks.

Furthermore, RMs for PSS are primarily determined by the degree of integration between the provider and customer. The higher the level of integration among these three roles, the greater the potential for creating sustainable value (Yang & Evans, 2019). In product-oriented PSS, the provider acts as the producer and customer as the owner or user (Tukker, 2004; Barquet et al., 2011); revenues primarily stem from the sale of physical products (Yang & Evans, 2019; Bonnemeler et al., 2010), also defined as asset sale (Osterwalder & Pigneur, 2010). Therefore, the financial risk for the provider is low but for customer it becomes a huge financial investment with lower dependency on provider (Yang & Evans, 2019). User-oriented PSS, the provider is both the producer and owner, and the customer is the user (Tukker, 2004; Barquet et al., 2011; Yang & Evans, 2019), enabling revenues mainly through leasing (Yang & Evans, 2019; Bonnemeier et al., 2010). Here, the financial risk for the provider is higher as they own the product, but the risk is low for customer as they avoid huge financial investments (Yang & Evans, 2019). In contrast result-oriented PSS, the provider holds all three roles (producer, owner, and user), providing full control over the lifecycle and allowing revenues to be based entirely on the delivered functional result. Hence, the customer pays for the result of the service which imposes a high risk for the provider, but at the same time customer is service user and depends on external technologies without any control and may face potential limitation in customization (Yang & Evans, 2019; Tukker, 2004). Consequently, the selection of revenue goes beyond pricing mechanism (Frohmann, 2023), it also determines how risk is allocated, how customer relationships are structured and how value is recognized and captured (Zott & Amit, 2010; Tabares et al., 2023).

In addition, for firms holding a patent – a form of technological protection – licensing becomes the central revenue (Heiden & Bereuter, 2022). Licensing involves share both success and risk, whereby the licensor grants access to technology in exchange for financial compensation (Heiden & Bereuter, 2022) and can be regarded as a type of revenue stream (Osterwalder & Pigneur, 2010). It enables firms to generate income from their technology through independent manufacturers, while allowing smaller firms to avoid substantial investments in their own facilities, thereby reducing both risk and cash flow requirements (Teece, 1986). However, a significant drawback of licensing is the loss of autonomy over technological development (Heiden & Bereuter, 2022; Holgersson & Van Santen, 2018), as well as the risk that the licensee imitates the technology and eventually begins competing directly with the licensor (Teece, 1986). Table 1 and 2 summarize how a firm can generate

revenues based on the abovementioned offerings, and their advantages and disadvantages from the provider and customers point of view.

Table 1: Comparison of revenue models from the provider's (DTV) point of view.

| | | Role | Main Revenue | Advantages | Disadvantages |
|---|---|---|---|---|---|
| PSS architecture | Product-oriented | Producer | Asset Sale | • Low capital commitment.<br>• Low risk. | • Limited differentiation.<br>• Low customer interaction. |
| | User-oriented | Producer and owner | Leasing | • Enables recurring revenue stream.<br>• Strengthens customer relation.<br>• Lower adoption barriers. | • High capital commitment.<br>• Higher risk.<br>• Difficult for small ventures. |
| | Result-oriented | Producer, owner and user | Pay per Result | • Strong relation with customer, that can build trust.<br>• Low adoption barriers. | • Carries the risk.<br>• Exposure to uncertainties in technology performance, operations, and external factors. |
| Technology | Patent | Licensor | License | • Enables sharing of success and risk between licensor and licensee.<br>• Lowers financial risk and cash flow pressure. | • Limited control of the technology.<br>• Risk that the licensee imitates the technology.<br>• Potential long-term weakening of competitive advantage. |

Table 2: Comparison of revenue models from the customer point of view.

| | | Role | Advantages | Disadvantages |
|---|---|---|---|---|
| PSS architecture | Product-oriented | Owner and user | • Full ownership and control.<br>• Low dependency on provider. | • High upfront investment.<br>• Carries the operational and technological risk. |
| | User-oriented | User | • Low adoption barriers.<br>• Low upfront investments. | • Dependency on provider.<br>• Long term recurring payment. |

| | | | | |
|---|---|---|---|---|
| | | | • Provider responsible for maintenance. | |
| | Result-oriented | Service user | • Pays for the outcomes.<br>• Low adoption barriers.<br>• Low operational responsibility. | • Limited control over process.<br>• Dependent on provider. |
| Technology | Patent | Licensee | • Access to advanced technology<br>• Control over the process | • Dependent on licensor. |

The implementation of revenue models goes beyond the pricing structure, as revenue models define the distribution of ownership, risk as well as operational responsibilities between customer and provider (Zott & Amit, 2010; Tabares et al., 2023). Therefore, firms need to compare revenue models and their influence on profitability, scalability, and customer-related requirements, and since different revenue models differ in terms of ownership operational responsibility and risk and investment allocation, firms need more a systematic approaches for comparing how these revenue models fit provider and customer requirements (Yang & Evans, 2019; Linde et al., 2023). Creating a demand for tools that can compare and support revenue model implementation from both provider and customer perspectives.

### 2.4 Axiomatic Design

As the implementation of revenue models involves balancing several provider and customer requirements, a structured design methodology was needed to support the development of the simulation-based prototype. According to Suh (2001), AD is a design theory created to support systematic problem solving by transforming functional requirements into implementable solutions. The theory is based on the concept that that complex systems can be structured by connecting functional requirements (FRs) with design parameters (DPs), helping designers translate conceptual needs into implementable and practical solutions (Suh, 2001). This makes AD well suited for the study, where multiple implementation-related requirements connected to revenue models needed to be systematically mapped, compared and integrated into simulation functionality.

The framework is structured around four domains: the customer domain, functional domain, physical domain and process domain (Suh, 2001). The customer domain focuses on the overall needs and expectations of the system, while the functional domain transfers these into functional requirements (FRs). The physical domain includes the DPs used to achieve FRs, while the process domain translates these DPs into implementable process

variables (PVs). Through this mapping process, AD enables a systematic breakdown of complex systems from abstract requirements into practical solutions (Suh, 2001). In this study, the FRs describe the core capabilities of the prototype, such as structure on revenue models and evaluate profitability and risk. The DP refers to the structural elements required to satisfy these FRs.

The framework builds upon two fundamental principles: the Independence Axiom and the Information Axiom (Suh, 2001). The independent Axioms states that FRs should remain mutually independent, ensuring that changes in one function should not negatively influence another. The Information Axiom suggests that the most desirable solution among viable solutions is the alternative characterized by lowest complexity and the highest probability of successful implementation (Suh, 2001). Together, these principles enable a structured and transparent design process while minimizing unnecessary complexity. However, a limitation of AD is that the framework can be difficult to apply in context with high level of uncertainty and ambiguous relationships, since defining independent FRs and clear relationship between FRs and DPs can be challenging (Suh, 2001).

AD provides a relevant framework for this study as the implementation of revenue model involves several factors such profitability, scalability and risk distribution as well as operational responsibility. By structuring these relationships into FR-DP mappings, AD provides a structured foundation for the modular design process for developing the simulation-based prototype.

### 2.5 Synthesis

The literature argues that DTVs are characterized by advanced technological offerings often involving integrated products, services, technical support, and long-term operational responsibilities (Raff et al., 2024; Romme et al., 2023). These characteristics create implementation conditions under which firms need to define how value is created, delivered and monetized between providers and customers. The literature identifies PSS as a relevant framework for understanding how these values can be structured and monetized for integrated offerings that structure ownership, operational responsibility and risk through different offering architectures (Tukker, 2004; Mont, 2002). Consequently, different PSS structures create opportunities for different revenue models, such as asset sales, leasing, licensing and result-oriented revenues which differ regarding profitability, scalability, ownership, and risk-sharing between providers and customers (Yang & Evans, 2019; Barquet et al., 2011).

Concurrently, existing literature suggests that revenue model implementation is increasingly challenging in DTV context due to the lack of clear market benchmarks as well as evolving implementation conditions (Raff et al., 2024; Romme et al., 2023). Different revenue models create distinct requirements for provider and customer regarding investments,

operational responsibilities and adoption. However, existing literature mainly addresses revenue structures, while offering limited understanding of how firms can compare and implement these models.

Building upon this, this study approaches revenue model implementation is viewed as a complex and context-dependent decision-making process influenced by DTV characteristics, PSS structures and provider–customer requirements. Therefore, the study identifies a need for more systematic approaches to support the comparison and implementation of revenue models. To address this complexity, the study applies AD as a structured design method to develop a simulation-based prototype. Through FR-DP mappings, AD offers a structured approach for translating implementation-related requirements into modular prototype functions. Therefore, the theoretical synthesis provides the analytical framework for understanding how a simulations-based prototype may support the comparison and implementation of revenue models within DTV contexts. Figure 1 therefore presents the theoretical synthesis and analytical framework guiding the development of the simulation-based prototype.

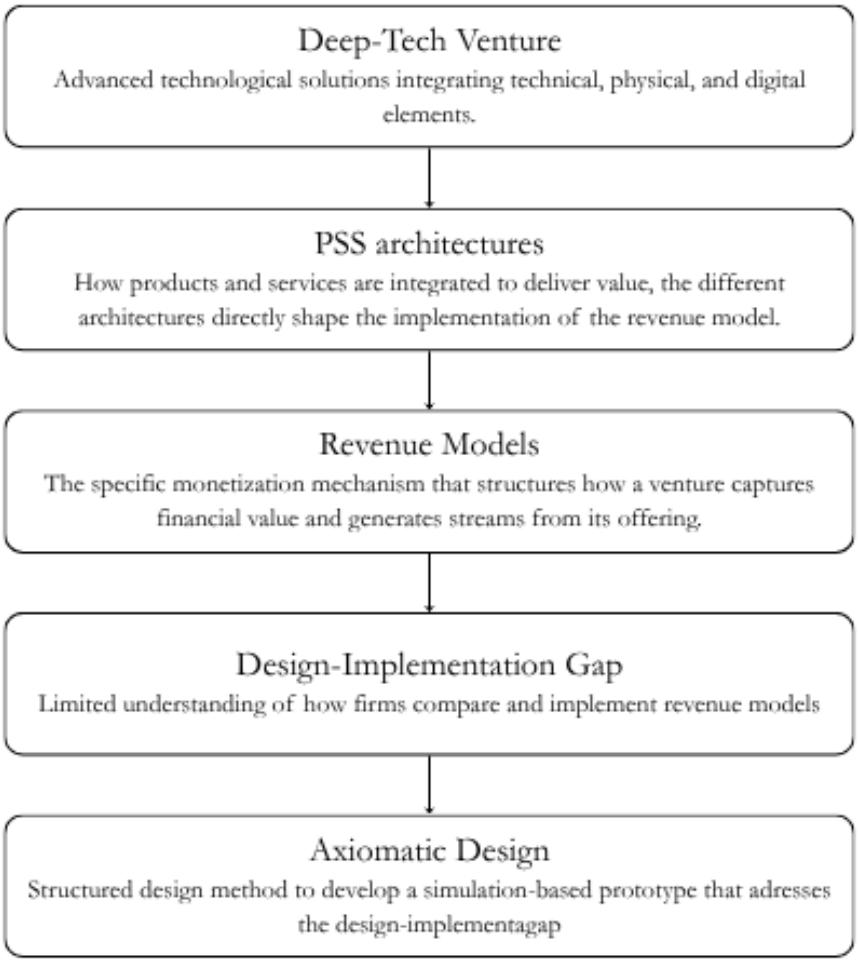



Figure 1: Theoretical framework for prototype development.

# 3 Methodology

*This section presents how the study was conducted adopting a qualitative case study approach based on semi-structured interviews with representatives from industrial customers and the case company. The collected empirical data was further used to for the development of the simulation-based prototype using axiomatic design.*

## 3.1 Research Approach

This study aims to investigate how a simulation-based prototype can be designed to support DTVs in the implementation of revenue models by integrating both organizational and customer insights. The research specifically focuses on the inherent challenges associated with the implementation phase of revenue models. To address this problem, a prototype was developed as the primary research artifact.

Because this study pursues the mission of solving a real-world problem – namely, the difficulties surrounding the implementation of revenue models – the objective is to develop an artifact in the form of a simulator. To engineer a solution to this identified problem, a design science research approach was applied. Hevner et al. (2004) emphasize that design science is utilized to solve practical, complex, real-world problems through the creation of artifacts. While artifacts can encompass frameworks, concepts, and other abstract research results, prototype development serves as a critical pathway to achieving practical impact (Romme and Holmström, 2023). In this study, design science is leveraged to address the specific challenges tied to revenue model implementation within DTVs. Prior research indicates that a gap often exists between the conceptual design of revenue models and their practical implementation. The simulation-based prototype serves as the central artifact developed to visualize and simulate various revenue models based on both firm and customer insights. To guide the development of this artifact, AD theory was employed. AD theory facilitated the infusion of a user-centric approach, systematically integrating customer insights into the development process.

Moreover, to foster a comprehensive understanding of customer and organizational insights, qualitative data was collected. Given that the analysis is centred on the organizational level, the empirical setting of the study is grounded in a qualitative research approach. This alignment complies with Taherdoost (2022), who argues that case studies are commonly based on contextual issues, practical problems, or lessons learned, and should therefore be regarded as a form of qualitative research. The empirical material was collected through semi-structured interviews with customers, supplemented by observational notes from consultations and meetings with the case company. This qualitative data was subsequently utilized to identify critical needs, challenges, and preferences, serving as the foundational baseline for the development of the artifact.

### 3.2 Data Collection

The primary data for this study was collected through an initial consultation and meetings with the case company, alongside semi-structured interviews with both their existing and potential customers.

The empirical process initiated with a consultation with the case. During this session, the Chief Executive Officer (CEO) and Chief Operating Officer (COO) of the case company provided vital information regarding the technology, as well as strategic input to ensure the artifact was aligned with organizational needs. This consultation lasted approximately two hours, during which the dialogue between the industry respondents and the research team was documented through detailed written notes.

Throughout the course of the project, regular alignment and follow-up meetings were conducted with the case company via Microsoft Teams. These internal sessions served multiple purposes. Initial meetings focused on scoping the project and mapping the radical technology to establish the foundation for the design process. Subsequent follow-up meetings were used to review the progress of the prototype, verify the underlying assumptions of the simulator, and calibrate the parameters based on the company's expertise. Data during each session were documented through detailed written notes. A summary of the company-side empirical material, including the roles of the participants, the purpose of each session, and the total duration, is outlined in Table 3 below.

Table 3: List of internal alignment consultations with the case company.

| Session | Participants | Purpose and Core Focus | Format | Date | Duration |
|---|---|---|---|---|---|
| Initial consultation | CEO & COO | Information regarding the technology, the company's problem, scoping the project, identifying organizational expectations | On-site | November 21st, 2025 | Around 120 min |
| Follow-up Alignment Meetings | CEO & COO | Expectation and requirements for the simulation | Digital | February 5th, 2026 | Around 50 min |

| | | | | | |
|---|---|---|---|---|---|
| Follow-up Alignment Meetings | CEO | Presenting different revenue models. Discussion about different revenue models, their risk and technology maturity. | Digital | March 20th, 2026 | Around 90 min |
| Follow-up Alignment Meetings | CEO | Values of input variables for the prototype. Discussion about their customers. | Digital | April 15th, 2026 | Around 80 min |

Furthermore, semi-structured interviews were conducted with existing and potential customers, as they are crucial in the evaluation and adoption of revenue models for the case company. According to Yin (1984), semi-structured interviews enable flexibility with a structured focus on important themes. David and Sutton (2016) further add that semi-structured interviews balance structure with flexibility, allowing follow-up questions and spontaneous elements, enabling exploration of emerging themes. Considering deep tech ventures involve limited historical data, semi-structured interviews are considered an effective approach to capture a nuanced insight about customer preferences regarding the case company's revenue model. The selection of respondents was done using a purposive sampling approach, targeting relevant individuals with relevance to this thesis. Selecting relevant respondents is crucial, as purposive sampling ensures that respondents possess the necessary knowledge and experience to provide valuable insights, strengthening the credibility and applicability of the findings (Bryman, 2016). The respondents represented diverse professional positions within their respective organizations; however, each had some varying level of involvement in process associated with evaluation, adoption and implementation of new technologies. Ensuring, that the data reflected multiple viewpoints while preserving consistency with the study's focus on choice of revenue model.

Each interviews had a span of 35-50 minutes and were conducted digitally through Microsoft Teams, depending on the respondent's availability and preference. The semi-structured approach allowed to follow an interview guide while allowing to explore emerging themes more thoroughly. Three respondents from three different companies were selected to conduct interviews mentioned in Table 4.

Table 4: List of customer respondents and details of interviews.

| Respondents | Role | Duration (min) | Date | Represented firm size | Industry |
|---|---|---|---|---|---|
| R1 | Senior Manager product pre-development | 41:21 | April 1st, 2026 | Mid-size | A |
| R2 | Specialist/material engineer | 47:57 | April 7th, 2026 | Startup | B |
| R3 | Material & technology development manager | 36:17 | April 10th, 2026 | Large | C |

The respondents were chosen according to their position within their organizations, primarily involving decisions-makers including managers, technical experts or individuals involved in assessing new technologies. Their different roles allowed them to provide important insights into how perceived value, risk, capital investment affect decision-making regarding revenue models decision.

All interviews were audio-recorded, with respondents' consent, to ensure a precise and reliable data collection. The recordings were subsequently transcribed and used as the basis for qualitative analysis. Interviewing participants previously unknown may pose challenges regarding openness and depth, however, these risks were minimized through clear communication of the study's purpose and assuring confidentiality, as advised by Collis & Hussey (2014).

### 3.3 Data Analysis

The empirical data collected from the consultation and meetings with the case company were analysed deductively, focusing heavily on the organization's needs, requirements, and assumptions regarding the prototype. Unlike the customer interviews, which were utilized to capture the consumer perspective, the notes were used to identify the corporate perspective and the FRs the artifact needed to fulfil. Consequently, this analysis concentrated on determining the economic assumptions that required integration, and the specific strategic decisions the prototype was intended to support.

The empirical data collected from customers was analysed using deductive coding inspired by the framework proposed by Fereday and Muir-Cochrane (2006). Deductive coding is a method within qualitative thematic analysis where researchers utilize pre-determined codes

to organize and interpret data (Fereday & Muir-Cochrane, 2006). Because the coding aimed to uncover design-relevant insights, these pre-determined codes were applied to operationalize both the customer and organizational perspectives within the artifact. This method was selected to identify implementation-related insight relevant that could directly inform the development of the prototype.

The methodology outlined by Fereday and Muir-Cochrane (2006) describes how a coding framework can be developed prior to the analysis – based on the research aim and theoretical framework – and subsequently applied to interview transcripts. In this study, the coding framework was developed based on the study's theoretical framework as well as the customer-related parameters destined for operationalization within the prototype. In alignment with Fereday and Muir-Cochrane's (2006) recommendations for codebooks, each code was explicitly defined through a code name, a conceptual definition, and a description of when the code should be applied.

This analysis was executed in several consecutive stages. First, the interviews were read thoroughly and summarized to establish a comprehensive overview of each respondent's perspective. Next, the coding framework was applied to the transcripts by highlighting relevant text segments and linking them to the predefined codes. While the coding process was primarily deductive, inductive codes were supplementary added when the empirical material revealed design-relevant insights that were not captured by the initial framework. Following the coding phase, the codes were clustered into design insights to be utilized in the development of the prototype. In this manner, the analysis served to translate customer insights into relative scenario values within the prototype.

To enhance research transparency, the linkages between the empirical excerpts, code categories, and simulator parameters were rigorously documented. Crucially, the defined parameter values should not be interpreted as statistically generalizable metrics; rather, they represent empirically informed design values utilized to integrate the customer perspective into the artifact.

Given that this study relies on a bounded empirical dataset, the objective was not to identify generalizable patterns, but rather to foster an understanding of various preferences and needs associated with the implementation of revenue models. Therefore, the analysis focused on interpreting and synthesizing the insights that emerged from the empirical material, examining how these could serve as a foundational baseline for the iterative development of the prototype.

### 3.4 Translation from qualitative insights to values in the Prototype

To bridge the gap between the qualitative empirical findings and the quantitative nature of the simulation-based prototype, a structured translation process (parameterization) was conducted.

First, the coding of the customer interviews identified customers preferred revenue model and precived risk. These qualitative themes were mapped against the functional requirements (FRs) and design parameters (DPs) established through the axiomatic design framework. To insert these insights into the simulator, qualitative statements were operationalized into numerical inputs or conditional logic. For instance, customer expressions of "high risk" were translated into specific risk-coefficient scoring scales that penalize purely transactional models within the simulation framework. Following scoring scale, ranging from one to four, was developed:

- Score one (Very low): Implies that a specific revenue model is highly undesired by the customer, or that an evaluated risk factor is perceived as negligible.
- Score two (Low): Indicates minor strategic preference or a low, manageable risk barrier.
- Score three (High): Reflects a strong strategic alignment/preference, or a substantial risk concern that acts as a significant barrier to implementation.
- Score four (Very high): Represents either an absolute preference or a critical, deal-breaking risk factor.

### 3.5 Design process of the Prototype with AD

To structure the development of the prototype, axiomatic design (AD) was employed as the foundational design theoretical framework. This specific theory was chosen because it enables a systematic translation of identified needs into FRs and, subsequently, into concrete DPs (Suh & Suh, 2001). In this study, AD was utilized to ensure that the prototype's functionalities were rigorously mapped to the research objective, the theoretical framework, and the empirical insights derived from both customer interviews and consultations with the case company.

Based on the determination of system requirements during the initial consultation with the case company, the identified needs were further elaborated. These collected needs were then used to define the FRs, which describe the intended actions and capabilities of the simulation. In this context, the FRs explicitly dictated what the prototype needed to accomplish to support the implementation of revenue models. These requirements were formulated with a balanced focus on both economic and customer-related dimensions, aligning with the study's aim to investigate how a simulation-based prototype can be designed to support implementation of revenue model for a DTV. For each identified FR, a specific DP was selected to represent the technical or structural solution. Within this architecture, DPs were operationalized as specific system modules. At the foundational level, the three main FRs were directly linked to three overarching modules.

To transition from abstract requirements to concrete functions, a "top-down" decomposition across domains was applied (Suh & Suh, 2001). Specifically, when a FRs was broken down into sub-requirements, the corresponding sub-design parameters were

defined immediately. These parameters were collaboratively elaborated within the research team, heavily informed by the collected empirical data.

Furthermore, to ensure the correctness of this mapping, design matrices were utilized to analyse how each DP affected its corresponding FRs. Each FR-DP pair was designed and rigorously evaluated against the independence axiom (Suh, 2001; Suh, 1998). Through the application of these design matrices, the research team ensured that each DP was either completely independent or that they adhered to a sequential dependency chain – known as a decoupled design (Suh & Suh, 2001; Suh, 1998) – thereby minimizing developmental complexity. The ultimate objective was to achieve an independent design where each module serves a distinct, isolated purpose. Subsequently, process variables (PVs) were defined by the research team to ensure that the architectural design of the prototype could be effectively implemented within Microsoft Excel for testing. Microsoft Excel is highly suitable for early-stage prototype development, scenario analysis, and the visualization of computationally driven models. Spreadsheets are frequently utilized as foundational systems where data, calculations, and visualizations converge to robustly support managerial decision-making (Tyran, 2010; Seila, 2004). Consequently, Microsoft Excel was deemed appropriate for this study because the prototype was required to facilitate dynamic comparisons between various revenue models and demonstrate how economic and customer-related parameters interactively influence outcomes. Because the primary objective was to develop a functional prototype rather than a fully commercialized software system, the study necessitated a tool that was accessible, highly malleable, and intuitive to further develop. Excel thus provided a flexible and practically viable prototyping environment capable of operationalizing the process variables identified through the AD process. The design process of the prototype is visualized in Figure 2, below.

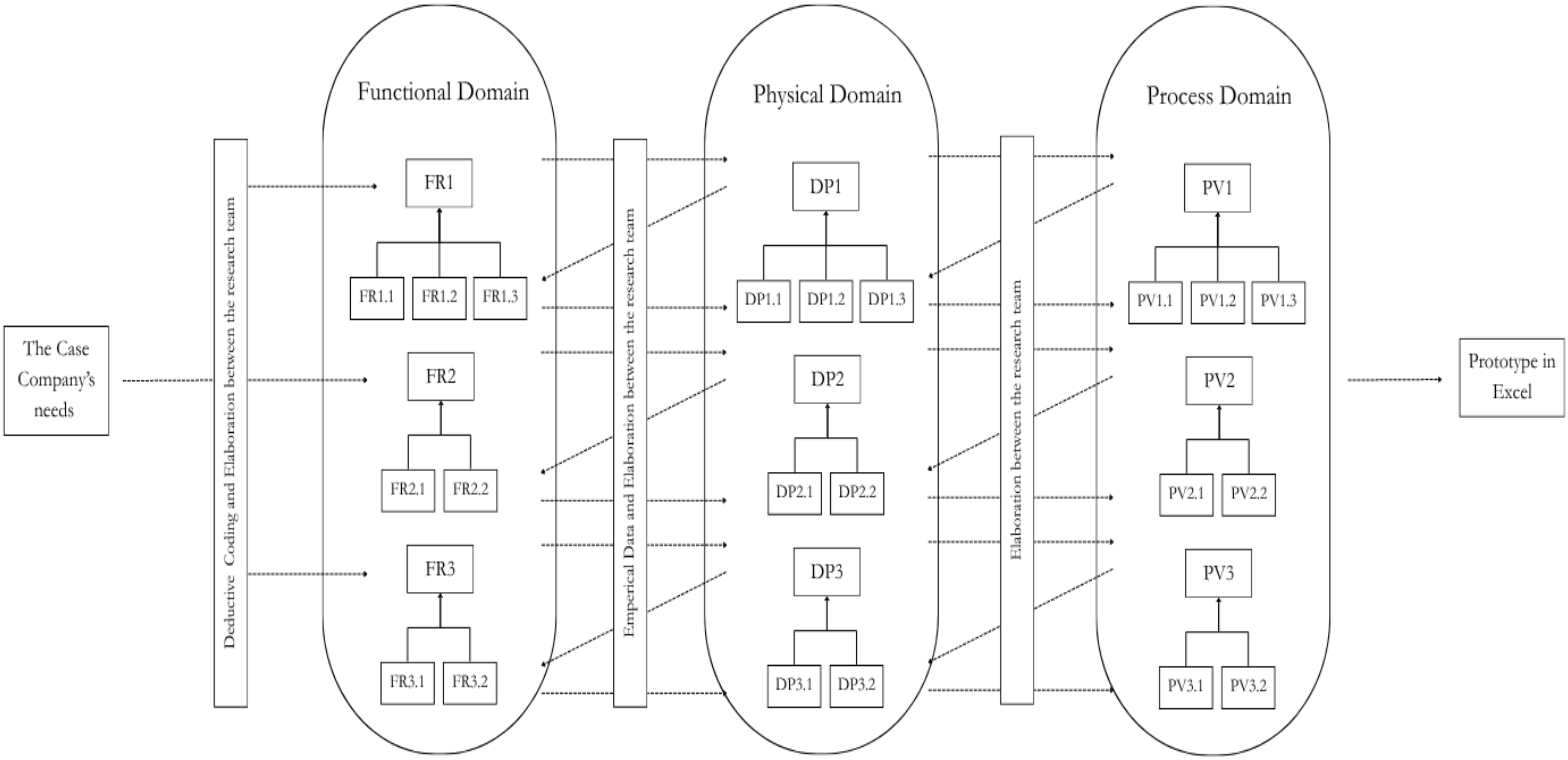



Figure 2: Visualization of the design development of the prototype using AD; mapping the transition from empirical needs for the case company to FRs, DPs, and process variables using a "top-down" decomposition within the different domains according to Suh & Suh (2001).

### 3.6 Ethical Consideration

When conducting qualitative interviews, it necessitates careful consideration of multiple ethical aspects to protect both respondents and the organizations they represent. As pointed out by Kvale (2009), the disclosure of confidential information may lead to participants become exposed to risk. Therefore, all participants and their respective companies have been anonymized using coded identifiers (R1, R2 and R3.), ensuring the identities of the individuals and organisations remain undisclosed.

Participation in the study was completely voluntary, and all the respondents were informed about the purpose of the study and their role in it. Before the interviews, participants were informed about the recording, and all the respondents gave their consent to be recorded. All the interviews were conducted digitally, and all recordings were handled in secure manner.

To further ensure the integrity of the participants, no identifiable information such as names, gender, or exact company details have been included in the analysis. This reflects the recommendations of David and Sutton (2016), who point out confidentiality and anonymization as fundamental ethical components of qualitative research. Furthermore, all collected data has only been used for purposes of this study and handled in accordance with General Data Protection Regulation (GDPR). The data will be deleted following the completion of the study.

### 3.7 Trustworthiness

To ensure the validity and reliability of the study, the four criteria introduced by Guba (1981) - credibility, transferability, dependability, and confirmability – are systematically applied. These criteria provide a systematic framework for assessing the trustworthiness of the research (Guba, 1981).

Credibility refers to how convincingly the findings capture and represent the phenomenon studied (Guba, 1981). Credibility in this study was improved by gathering data from several respondents representing various companies, and company sizes. These respondents possessed many years of experience working and engaging with multiple new technologies and had substantial experience of their implementation processes. The variation mentioned above enabled the identification of recurring patterns across different context rather than relying on single perspective alone. Furthermore, the semi-structured interviews enabled the respondents to elaborate the on their experiences, resulting in detailed and insightful data. The integration of direct quotations in the findings increases credibility by clearly connecting interpretation to empirical data.

Transferability addresses to what extent the results are applicable to contexts (Gray, 2017). Although this is based on a specific case company, attempts were made to interview respondents from diverse organizational backgrounds, including both small startups and major industrial companies. By providing comprehensive description of the research context, data collection process, and respondent profiles, the study enables readers to assess the transferability of the results to comparable deep-tech and industrial context.

Dependability addresses the reliability and consistency of the research process (Guba, 1981). In qualitative research, the researcher acts as the primary instrument, allowing a flexible approach as new insights arise (Guba, 1981). To ensure dependability, the research process for data collection and data analysis have been described in detail. Consistency across interviews was maintained using the interview guide, while still allowing flexibility to explore themes that emerged during discussions.

Finally, confirmability addresses the degree to which the findings are based on empirical evidence rather than influenced by researcher's bias (Lincoln & Guba, 1985). This was addressed by relying on empirically grounded analysis through systematic coding and validating the findings with direct quotations from respondents. Moreover, the combination of inductive and deductive coding enabled a balance between theoretical guidance with openness to emerging insights, contributing to study's objectivity.

# 4 Development and Result of the Prototype

*This section presents the findings from empirical data and the development of the simulation-based prototype. This section details how the case company's needs and customer insights were systematically translated into functional requirements using axiomatic design, resulting in a Excel-based simulation prototype that evaluates and compares different revenue models.*

## 4.1 The Case Company's needs translated to first level FRs and DPs

Based on the determination of system requirements in the initial consultation och meetings with the case company, the needs were elaborated. These meetings served as a foundational source for requirements gathering, providing critical contextual understanding necessary for the development of the artifact. Consequently, these identified needs were utilized to define the FRs that describe the intended actions and capabilities of the prototype. Following summary of needs was elaborated (see Figure 3).

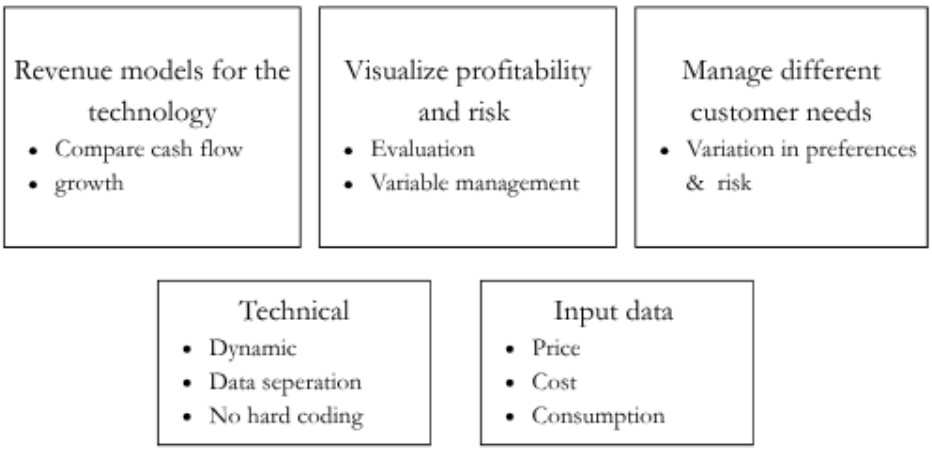



Figure 3: Collected and clustered needs from the case company.

The collected needs were used to define the FRs for the prototype development. The results in Figure 2 were analysed in order to define which inputs are constraints and which ones are FRs. The needs "Technical" and "Input data" were considered constrain in the axiomatic design process as they specified how the prototype should be implemented and specified which types of input data the prototype needed to handle. These constraints limited and guided the design of the prototype.

The needs – involving "Revenue model for the technology", "Visualize profitability and risk" and "Manage different customer needs" – were used to define the FRs and related DPs for the prototype development. The following first level of FRs and related DPs were defined:

- $FR_1$: Structure on revenue models
- $FR_2$: Evaluate profitability and risk
- $FR_3$: Customer variation

- $DP_1$: Revenue models module
- $DP_2$: Calculation and simulation module
- $DP_3$: Customer profile module

The following design matrix of the first level shows an independence design between the solutions (DPs) and the FRs:

$$\begin{Bmatrix} FR_1 \\ FR_2 \\ FR_3 \end{Bmatrix} = \begin{bmatrix} X & 0 & 0 \\ 0 & X & 0 \\ 0 & 0 & X \end{bmatrix} \begin{Bmatrix} DP_1 \\ DP_2 \\ DP_3 \end{Bmatrix} \quad (1)$$

### 4.2 Findings of Customer Insights

The deductive coding of the customer interviews was conducted based on two predefined parameters: preferred revenue model and perceived risk. These parameters were selected because they were to be operationalized within the prototype to integrate the customer insights into the comparison of different revenue models. The results are presented according to each respective parameter, demonstrating how the customers' insights served as the empirical foundation for the prototype's scenario values (described more in details in *4.3.3*). Given that the data consists of three interviews, the findings should not be interpreted as statistically generalizable, but rather as customer-specific insights relevant to the design of the artifact. The coding of the interview material has been compiled into a table (Table 5) to support the structure of the analysis.

Table 5: Coding of customer interviews.

| Parameters | Category | Code |
|---|---|---|
| Preferred revenue model | Product-oriented | purchasing of machinery, in-house, ownership, investment in equipment, exclusivity linked to product/technology |
| | User-oriented | leasing, renting, access to machinery |
| | Result-oriented | service model, treatment-as-a-service, payment per treated component, outcome/function |
| | Licensing of patent | licensing, patent, technology |
| Perceived risk | Technical maturity and risk | need for testing, uncertain functionality, technical validation |
| | Investment risk | high initial cost, purchasing machinery before demand is secured |
| | Certification and quality risk | certification, quality systems, auditing, standards, customer/regulatory requirements |

### 4.2.1 *Preferred revenue model*

The parameter preferred revenue model refers to the customer's preferred way of accessing, paying for or implementing the technology. The coding showed that customer preferences varied depending on factors such as technology maturity, internal capacity, demand uncertainty and willingness to commit to ownership or long-term investment.

In customer case R1, the preferred revenue model was strongly phase dependent. Initially, the respondent expressed a preference for a result-oriented service model, stating that

*"We usually start with a service model always at the beginning […] to cover the time period until we have the machine in-house or until we see the need for buying a machine, because the capacity demand is so high".* (R1)

This indicates that result-oriented arrangements are perceived as suitable in early stages when the customer wants to test the technology and gather market feedback before making larger commitments.

However, R1 also expressed openness toward user-oriented revenue models, particularly leasing or pay-per-use arrangements. The respondent explained that ownership was not necessarily required:

*"No, we do not need to own. They can put the machine even in our facility […] we just pay them like a monthly payment, or we pay them per volume. It's like leasing a car."* (R1)

This suggests that user-oriented model can function as an intermediate alternative where the customer gains access to the technology without taking full ownership or investment responsibility.

Product-oriented models, such as purchasing and integrating the machine internally, became relevant only under specific conditions. R1 emphasized that larger investments would only become relevant if the technology proved commercially viable and capacity demand increased:

*"We will only do this in case we are sure that the machine is busy […] and in case our customer is also happy with the performance."* (R1)

In contrast to R1, R2 expressed a more long-term preference for a result-oriented model. The respondent stated that they did not foresee a situation where the case company would no longer be needed if the technology worked successfully:

*"I don't see a path forward where we would end up being okay, we don't need them [the case company] anymore if the technology actually works we would most likely have a service-based model with them."* (R2)

This preference was linked to both capacity limitations and dependence on external expertise. The respondent further explained that internalizing the treatment process was unrealistic from a capacity perspective:

*"We would probably have somebody to do all these service treatments as well as [the case company] for these components […] rather than trying to have it in the house because that is not realistic from our capacity anyway."* (R2)

In customer case R3, the preferred revenue model was less defined due to the early maturity stage of the technology. The respondent described the collaboration as still being in an evaluation phase:

*"At this point we are still at the too early stage […] we have made our test campaign […] but we have not had time at this point to analyze the results. […] If it works… then it might be part of our production line"* (R3)

Although the respondent suggested that the technology could potentially become integrated into production in the future the interview indicates that revenue model decisions are dependent on further testing and validation.

#### 4.2.2 *Perceived risk*

The parameter perceived risk refers to the customer's perceived uncertainty and potential barriers associated with different revenue models for the technology. The coding showed that perceived risk was primarily connected to technology maturity, investment uncertainty, testing requirements and dependence on external expertise.

In customer case R1, perceived risk was strongly linked to technological uncertainty and the possibility that the technology may fail even after investment. The respondent explicitly acknowledged the uncertainty associated with adopting a new technology, stating that:

*"It might happen that this technology might not work, and we might even find out after investment that it will not work. But this is something that's a risk."* (R1)

This suggests that customers perceive a significant risk in committing to more investment-intensive revenue models before the technology has been sufficiently validated.

Investment willingness in R1 was therefore directly connected to risk reduction through testing and validation. The respondent emphasized that investments would not be approved before extensive testing had been completed successfully:

*"Before that risk is not solved, we will not invest at all […] I will definitely not get any approval to invest […] before all the testing, including testing in the field, are completed and positive."* (R1)

This demonstrates that perceived risk affects both the willingness to invest and the attractiveness of different revenue models. More ownership-oriented models appear less relevant before technical uncertainty has been reduced.

In customer case R3, perceived risk was strongly shaped by the industrial context. The respondent emphasized that:

*"We work in the [specific] industry"* (R3)

This indicate that risk is influenced by the high safety, certification and quality requirements associated with the customer applications. The interview further showed that extensive testing would be required before implementation could be considered:

*"We would have to do extensive testing […] one thing is lab scale testing but then the next part would be […] more complex geometries."* (R3)

This suggests that perceived risk in R3 is not only technological, but also connected to qualification, certification and the complexity of scaling the technology from laboratory conditions to industrial application.

In customer case R2, perceived risk was linked to both competence dependency and financial uncertainty. The respondent expressed that the company lacked the expertise required to perform the treatment process internally, explaining that:

*"We would probably not have the knowledge that was needed […] [the case company] have built up over time to do as good a job."* (R2)

This indicates that the customer perceives additional risk in internalizing the technology due to limited process knowledge and operational capability. R2 also emphasized that testing new technologies inherently involves financial risk:

*"There will always be a financial risk because it will have costs involved in testing these different new technologies […] we would have to do extensive testing."* (R2)

Similar to R1 and R3, this demonstrates that testing and validation are viewed as necessary before larger commitments or implementation decisions can be made.

### 4.3 Decomposing and Mapping process of second level FRs and DPs

The decomposition process of first-level FRs and DPs to second level aims to transform the abstract requirements into more tangible parameters that are close to the daily practice and therefore relevant for implementation. The FR-DP pairs on first level are a starting point for the "top-down" decomposition and mapping process in AD. The decomposition is performed separately for each of the FR-DP to obtain a better understanding of the process.

### 4.3.1 *$FR_1$-$DP_1$ – Revenue models for the technology*

Revenue models for the technology resulted in the development of a modular revenue model structure in the prototype. The prototype was designed to separate revenue streams and cost structures into configurable components, allowing different combinations of revenue models to be compared. Therefore, a library structure where revenue streams and cost elements could be defined independently from specific revenue models. This enabled the prototype to represent different logics, including product-oriented, user-oriented, result-oriented and licensing revenue models. A variable structure where parameters such as price, cost and volume could be adjusted dynamically. This allowed the prototype to evaluate how changes in assumptions affect profitability and revenue generation. A model matrix where different combinations of revenue streams and cost structures could be configured into revenue models. This enabled comparisons between multiple configurations within the same simulator environment.

- $FR_{1.1}$: Identify which revenue streams and cost structure are included
- $FR_{1.2}$: Manage variables for respective revenue stream and cost structure
- $FR_{1.3}$: Define different revenue models

- $DP_{1.1}$: Library of revenue streams and cost structure
- $DP_{1.2}$: Variable structure per revenue stream and cost structure
- $DP_{1.3}$: Model matrix for combinations of revenue streams and cost structure

The following design matrix shows a decoupled design:

$$\begin{Bmatrix} FR_{1.1} \\ FR_{1.2} \\ FR_{1.3} \end{Bmatrix} = \begin{bmatrix} X & 0 & 0 \\ 0 & X & 0 \\ 0 & X & X \end{bmatrix} \begin{Bmatrix} DP_{1.1} \\ DP_{1.2} \\ DP_{1.3} \end{Bmatrix} \quad (2)$$

The design matrix indicates a decoupled design, where $FR_{1.1}$ and $FR_{1.2}$ can be satisfied independently, while $FR_{1.2}$ depends on both the variable structure and the model matrix. This structure was considered acceptable because revenue model combinations require the underlying variables to already be defined before simulation can be created.

### 4.3.2 *$FR_2$-$DP_2$ – Visualize profitability*

Visualize profitability resulted in the implementation of scenario simulation and visualization functions in the prototype. The purpose was to allow the user to compare revenue models under different assumptions and uncertainty levels. Therefore, a scenario simulation structure where uncertain input variables, such as demand, could be modified to evaluate how uncertainty affects financial outcomes. This enabled the prototype to represent different implementation conditions. Moreover, a dashboard structure containing graphs and financial comparison metrics. The dashboard was designed to visualize profitability, risk exposure and differences between revenue model scenarios in a more accessible manner.

- $FR_{2.1}$: Evaluate uncertainty in input variables
- $FR_{2.2}$: Visualize and compare revenue models

- $DP_{2.1}$: Scenario simulation
- $DP_{2.2}$: Dashboard with graphs and risk metrics

The following design matrix shows an uncoupled design:

$$\begin{Bmatrix} FR_{2.1} \\ FR_{2.2} \end{Bmatrix} = \begin{bmatrix} X & 0 \\ 0 & X \end{bmatrix} \begin{Bmatrix} DP_{2.1} \\ DP_{2.2} \end{Bmatrix} \quad (3)$$

The design matrix represents an uncoupled design, indicating that uncertainty evaluation and visualization can operate independently from each other. This improved the flexibility of the prototype and simplified adjustments during development.

### 4.3.3 *$FR_3$-$DP_3$ – Manage different customer needs*

Customer analysis resulted in the integration of customer-specific parameters into the prototype. The purpose was to represent variations in customer preferences and perceived risk across the different customers. Therefore, a customer library structure where different customers could be represented within the prototype. The customers were based on the coded interview results. Parameter settings – such as preferred revenue model and perceived risk – for each customer were translated into relative values that could influence the attractiveness of different revenue models within the prototype.

The integration of customer parameters made it possible to compare how the same revenue model could be perceived differently depending on customer context. For example, result-oriented revenue models received higher preference values in customer case where uncertainty was high and had limited internal capacity, while product-oriented models became more relevant in with higher technology maturity and greater demand certainty.

- $FR_{3.1}$: Represent different customer segments
- $FR_{3.2}$: Represent variation in preferences

- $DP_{3.1}$: Library of customers
- $DP_{3.2}$: Key parameters settings for each customer

The following design matrix shows an uncoupled design:

$$\begin{Bmatrix} FR_{3.1} \\ FR_{3.2} \end{Bmatrix} = \begin{bmatrix} X & 0 \\ 0 & X \end{bmatrix} \begin{Bmatrix} DP_{3.1} \\ DP_{3.2} \end{Bmatrix} \quad (4)$$

The design matrix represents an uncoupled design, where customer representation and parameter settings can be managed independently. This structure allowed customer profiles to be added or modified without requiring changes to the overall logic.

### 4.4 Mapping of first and second level PVs

To operationalize the DPs in Excel, the DPs were further mapped into process variables (PVs). The purpose of the PV mapping was to define how the DPs would be implemented practically within the prototype environment.

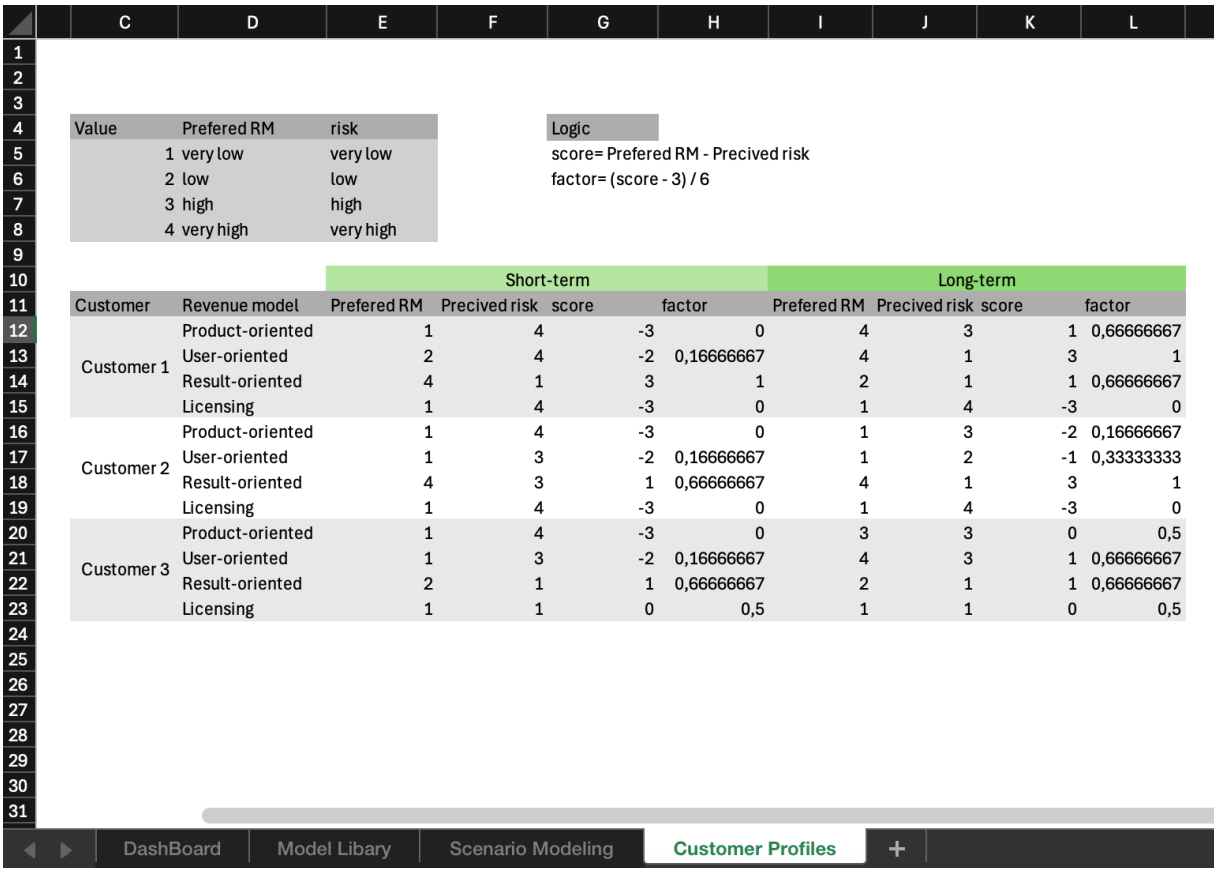
The PV mapping translated the abstract DPs into concrete prototype components. This made it possible to operationalize the revenue models, customer parameters and scenario simulations within the Excel prototype. The mapping also supported modularity by separating input variables, calculations and visualization components.

### 4.5 Summary of Prototype

The final prototype was developed in Excel (see Figure 4 to 7). The dashboard functions as the main interface of the prototype. In the dashboard, the user can select a specific customer, adjust input variables, and compare how different revenue models perform under different assumptions. The dashboard visualizes financial outcomes and customer-related fit, making it possible to assess not only which revenue model is financially attractive, but also which model is more aligned with customer preferences and perceived risk.

The customer profile sheet supports the dashboard by storing the scoring matrix for each customer. This matrix translates qualitative interview findings into numerical values that influence the prototype's evaluation of each revenue model. In this way, the prototype connects the empirical customer insights to the simulation logic.

The following figures (Figures 4 to 7) illustrate the customer profile sheet and the comparison of revenue models under different customer settings. Due to confidentiality considerations, detailed input values related to the case company's internal assumptions are not displayed in the Figures 5 to 7. The purpose of the figures is therefore to illustrate the prototype's structure, interface and output logic rather than disclose the underlying company-specific data.



| Value | Prefered RM | risk |
|---|---|---|
| 1 | very low | very low |
| 2 | low | low |
| 3 | high | high |
| 4 | very high | very high |

Logic

score= Prefered RM - Precived risk

factor= (score - 3) / 6

| | | Short-term | | | | Long-term | | | |
|---|---|---|---|---|---|---|---|---|---|
| Customer | Revenue model | Prefered RM | Precived risk | score | factor | Prefered RM | Precived risk | score | factor |
| Customer 1 | Product-oriented | 1 | 4 | -3 | 0 | 4 | 3 | 1 | 0,66666667 |
| | User-oriented | 2 | 4 | -2 | 0,16666667 | 4 | 1 | 3 | 1 |
| | Result-oriented | 4 | 1 | 3 | 1 | 2 | 1 | 1 | 0,66666667 |
| | Licensing | 1 | 4 | -3 | 0 | 1 | 4 | -3 | 0 |
| Customer 2 | Product-oriented | 1 | 4 | -3 | 0 | 1 | 3 | -2 | 0,16666667 |
| | User-oriented | 1 | 3 | -2 | 0,16666667 | 1 | 2 | -1 | 0,33333333 |
| | Result-oriented | 4 | 3 | 1 | 0,66666667 | 4 | 1 | 3 | 1 |
| | Licensing | 1 | 4 | -3 | 0 | 1 | 4 | -3 | 0 |
| Customer 3 | Product-oriented | 1 | 4 | -3 | 0 | 3 | 3 | 0 | 0,5 |
| | User-oriented | 1 | 3 | -2 | 0,16666667 | 4 | 3 | 1 | 0,66666667 |
| | Result-oriented | 2 | 1 | 1 | 0,66666667 | 2 | 1 | 1 | 0,66666667 |
| | Licensing | 1 | 1 | 0 | 0,5 | 1 | 1 | 0 | 0,5 |

DashBoard | Model Libary | Scenario Modeling | Customer Profiles

Figure 4: Customer profile sheet containing the scoring matrix for preferred revenue model and perceived risk.

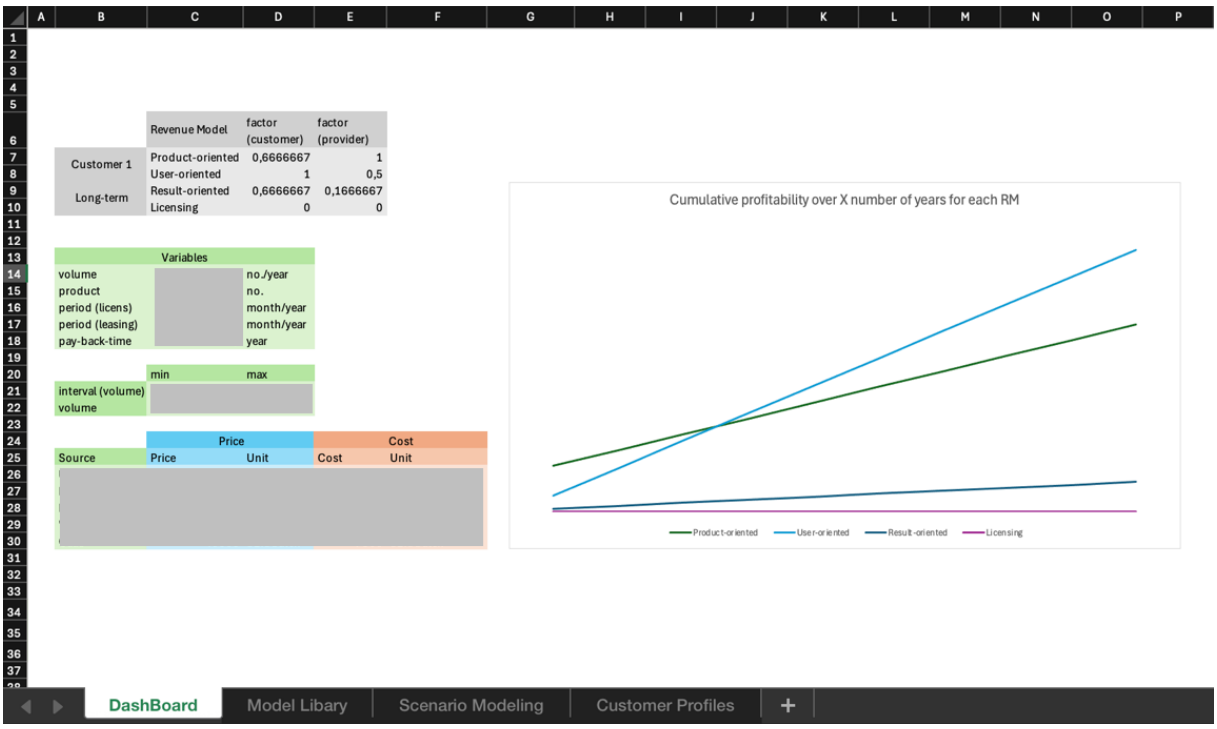



Figure 5: Excel dashboard showing comparison of revenue models for Customer 1.

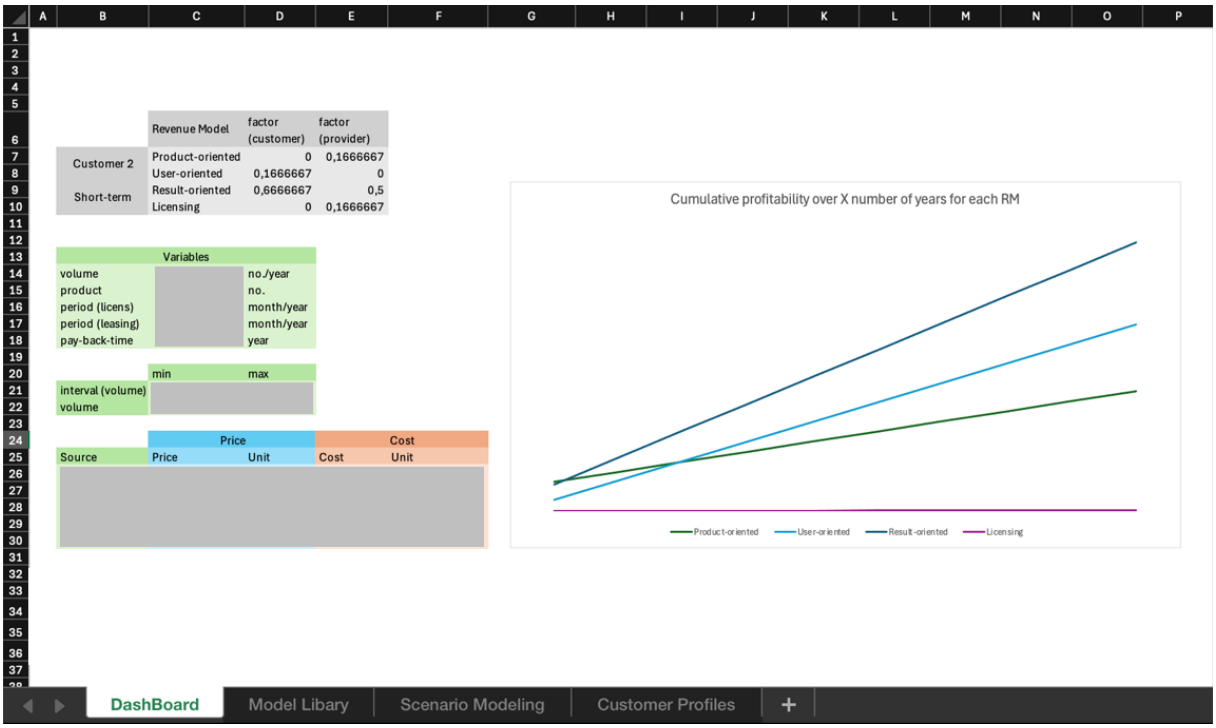



Figure 6: Excel dashboard showing comparison of revenue models for Customer 2.

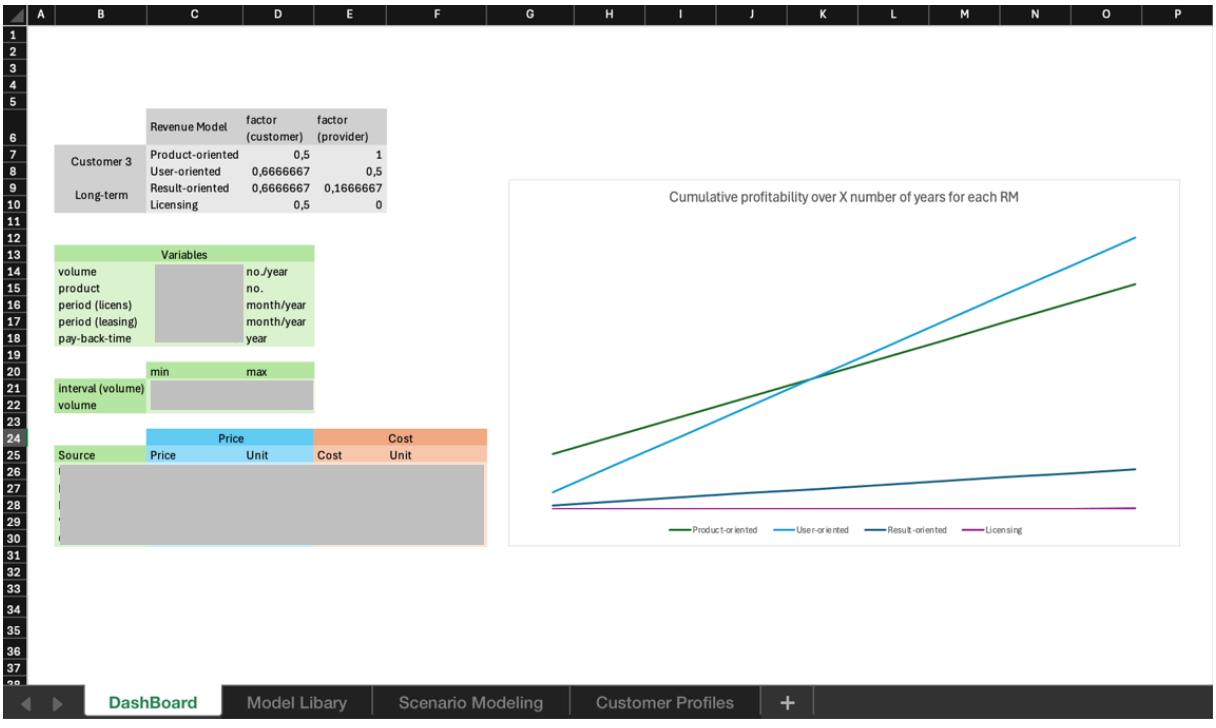



Figure 7: Excel dashboard showing comparison of revenue models for Customer 3.

# 5 Discussion

*This section contextualizes the findings in relation to existing literature and the study's purpose. Additionally, it presents the theoretical, and managerial implications, sustainability considerations limitation and future work.*

This study aimed to investigate how a simulation-based prototype can be designed to support the implementation of revenue models within DTVs through the integration of customer insights. Based on the findings, the research question: *How can a simulation-based prototype be designed to support the comparison of revenue models in DTV's from the provider perspective and insights from their customers?* – can be answered as follows.

The findings show that such a prototype can be designed to compare and adapt already identified revenue models in relation to both provider and customer preferences. In this way, the study addresses the design-implementation gap regarding revenue model implementation, where the challenge is to understand under which conditions they can be implemented.

The use of AD contributed to this by translating identified needs into FRs and DPs. By structuring the prototype into separate modules for revenue models, simulation calculations, and customer profiles, the design created a modular structure where different assumptions could be adjusted without disrupting the entire model. This supports the idea that AD can be useful when developing artefacts that need to handle complex and interdependent business model variables (Suh & Suh, 2001; Suh, 1998). However, the findings also show that AD has limitations. Customer preference and perceived risk are context-dependent and may change over time as the technology becomes more mature. Therefore, the mapping between FRs, DPs and process variables should not be understood as objective or final, but rather as an interpreted and simplified representation of the empirical material.

## 5.1 Theoretical Implications

The findings partly confirm previous literature on product-service systems and revenue model implementation. Revenue models are highly dependent on risk allocation, ownership, responsibility and customer relations (Tabares et al., 2023; Tukker, 2004; Barquet et al., 2011). Tukker (2015) and Mont (2002) argue that user-oriented and result-oriented models can reduce adoption barriers by lowering the customer's need for ownership and large upfront investments. Similarity, Yang & Evans (2019) emphasize that these models frequently require providers to assume higher financial and operational risks. This is reflected in the empirical material, where customers in early evaluation phases preferred models that reduced their exposure to technological and financial uncertainty. For example, result-oriented arrangements were perceived as attractive when the customer still lacked sufficient evidence regarding the technology's performance.

The findings also support Kowalkowski and Ulaga's (2017) argument that customers may depend on the provider's knowledge when adopting advanced technologies. This was particularly visible in cases where customers lacked the internal process knowledge required to operate or further develop the technology themselves. In such cases, customers did not only evaluate the revenue model in financial terms, but also in relation to the provider's ability to reduce operational uncertainty and support implementation. Which can be linked to Frohmann (2023) statement that revenue model goes beyond pricing.

At the same time, the findings challenge the assumption based on Tukker (2004), that more advanced result-oriented models are always strategically superior. In the context of DTVs, result-oriented models may reduce risk for the customer but increase risk for the provider. Since DTVs often operate with limited financial resources and high technological uncertainty (Raff et al., 2024), taking full responsibility for performance outcomes can create substantial exposure. This means that a model that appears attractive from the customer's perspective may be problematic from the company's perspective. The study therefore highlights the importance of analyzing revenue model fit as a balance between the customer and provider.

The findings also suggest that revenue model preferences are not static. Instead, customers may prefer different models depending on the maturity of the technology, the level of uncertainty and the expected production volume. For example, a result-oriented model may be preferred during early validation, while leasing or asset sale may become more relevant once the technology has been proven and demand has increased. This challenges static classifications of product-oriented, user-oriented and result-oriented models by showing that revenue model implementation can be phase-dependent and reversible.

Licensing also emerged as a theoretically relevant but practically complex revenue model. Teece (1986) describes licensing as a way for research-intensive firms to generate revenue from the patent without large investments in production capacity. This can be attractive for DTVs because it reduces the need for capital-intensive scaling. However, the findings show that licensing may be less attractive for customers when the technology is still uncertain and requires substantial process knowledge. In such cases, customers may prefer models where the supplier remains involved in implementation, operation or validation. This indicates that licensing may be more suitable at later stages, when the technology is better understood and the customer has sufficient internal capabilities to use it.

### 5.2 Managerial and Practical Implications

From a managerial perspective, the prototype should be understood as a decision-support and discussion tool rather than as an exact forecasting instrument. The simulator can help DTVs identify situations where a revenue model is financially attractive for the company

but perceived as risky or unattractive by the customer. This makes it possible to adjust the implementation strategy before committing to a specific model.

For managers in DTVs, the findings suggest that revenue model implementation should be approached as a staged process. In early phases, when uncertainty is high, result-oriented or pilot-based models may be useful because they allow customers to test the technology without committing to ownership or large investments. As the technology becomes more validated, the company may gradually introduce leasing, licensing or asset sales depending on customer demand, production volume and risk distribution.

The prototype can also support communication between the provider and the customer. By visualizing how different assumptions affect financial outcomes and adoption potential, the simulator can be used to explain why a certain revenue model may be more suitable in the long term, even if it is not initially preferred by the customer. In this way, the tool can support negotiation and alignment between the company's preferred revenue model and the customer's perceived risk.

However, the practical value of the simulator depends on how the input values are defined and updated. Since the customer parameters in this study are based on qualitative interviews, they should not be interpreted as statistically representative market data. Instead, they should be seen as empirically informed design values that demonstrate how customer insights can be translated into a simulation environment. The main contribution is therefore not the specific numerical values, but the structure that allows qualitative insights to be integrated into revenue model evaluation.

### 5.3 Sustainability Considerations

Sustainability is widely understood from a triple bottom line perspective, which integrates environmental, economic and social sustainability (Alhaddi, 2015). In this study, the discussion will focus on all three aspects, environmental, economic och social sustainability, since these dimensions of sustainability are closely linked to the technological offerings as well as simulation-based tool.

The advanced technology aims to reduce corrosion and hydrogen embrittlement in industrial applications, potentially extending product life cycle, reduced material waste and reduced need for maintenance. From an environmental sustainability perspective, extending the life cycle of industrial components can reduce resource consumption and improve long-term sustainability performance. In this regard, the technology can be seen as sustainable solution as it contribute to sustainability by reducing waste and extending product life cycle, particularly in industries where demanding operational environments often cause extensive material degradation and operational inefficiencies.

Furthermore, this study contributes to sustainable decision-making by developing a simulation-based prototype that is designed to assist firms and managers in comparison and implementation of several revenue models. From a social sustainability perspective, the simulation-based prototype may simplify complex decision- making by helping managers to visualize how various revenue models affect operational responsibility, investment, profitability and customer requirements. By structuring and comparing different revenue mode alternatives, the simulation-based tool may reduce uncertainty in strategic decision-making and improve collaboration between the provider and customer. This may improve organizational and managerial working conditions by reducing inefficient implementation processes, strengthening communication and enabling more transparent decision-making regarding long-term implementation strategies.

From an economic sustainability perspective, the simulation-based prototype can assist firm in comparing revenue models that enhance profitability, scalability and long-term value capture. By simulating how different revenue models affects revenues, operational costs, investments needs and risk distribution, firms can improve their ability to select economically sustainable implementations option. In this context, the simulation-based prototype may support improved utilization of organizational resources and reduce the risk of selecting and implementing unsuitable revenue models that may lead to financial and operational inefficiencies. Therefore, the study contributes to sustainability beyond the technological capabilities of technology, but also through the development of structured decision support tool that improves long-term operational and organizational sustainability within the DTV.

### 5.4 Limitations and Future work

This study has several limitations. First, the customer insights are based on three qualitative interviews. This means that the findings cannot be statistically generalized to all potential customers or industries. The customer profiles used in the simulator should therefore be understood as illustrative and empirically informed examples rather than representative market segments.

Second, the simulator is based on simplified assumptions regarding customer preferences and perceived risk. Although simplification is necessary in a prototype, it also means that the model does not capture all organizational, technical, regulatory and market-related factors that may influence revenue model implementation. The translation from qualitative interview data into quantitative simulator values also involves researcher interpretation, which may affect the results.

Third, the prototype has not yet been evaluated in real decision-making contexts by the company or by customers. Therefore, the study cannot determine whether the simulator improves actual revenue model decisions in practice. Future research should test the

prototype with deep-tech ventures and potential customers in real or simulated decision-making situations.

Finally, future studies could expand the simulator by including more customer parameters, such as technological maturity, regulatory requirements, internal capabilities and willingness to share risk. Additional interviews with a broader range of customers would also strengthen the empirical basis of the model. This would make it possible to further develop the simulator from a proof-of-concept into a more robust decision-support tool for revenue model implementation in deep-tech ventures.

# 6 Conclusions

*This chapter concludes the study by presenting the theoretical and practical contribution of the study focusing on revenue model comparison and implementation and the development of a simulation-based prototype.*

The study contributes theoretically by explaining how AD can be applied beyond its original engineering framework to support revenue model implementation in DTV contexts. By mapping implementation-related requirements into FRs and DPs, a modular framework is developed where individual assumptions can be modified without disrupting the overall logic. This suggests that AD is useful when complex and interconnected business model variables need to systematically structured. At the same time, the study underlines that the FR-DP mapping should be regarded as an analytical interpretation rather than a fixed or universal representation.

From a practical perspective, the study offers a simulation-based prototype designed to support decision-making within the company. The prototype enables firms to evaluate revenue models from providers perspective together with customers insights, helping firms to identify potential conflict areas before committing to a specific model. By integrating customer related parameters identified through qualitative interviews, the prototype can visualize why a certain revenue model may be appropriate for a customer in the long term, despite initial perceptions suggest otherwise. This support negotiation and alignment between the providers preferred revenue model with the customer's perceived risk. The prototype supports a phased implementation process, allowing firms to start with a pilot-based or result-oriented models and progressively shift toward leasing more investment intensive models as the technology matures and market demand increases.

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# Appendix – Interview Guide

### Introduction

- Present names and educational background and explain conducting our master thesis in collaboration with the case company.
- Brief explanation of purpose of the study.
- Clarifying that the interview will be recorded, that the interviews will be anonymous in the final report, and ask for consent about this. Explaining how the recording will be handled.
- Ask if they have any questions before starting.

### Background Questions

- Can you briefly describe your position in the company?
- Have you previously been involved in processes related to the implementation of new deep tech technologies?
    - If YES: How many? Briefly describe, in general terms, how the internal process for evaluating and implementing the technology was carried out.
    - If NO: *Proceed to the next question*

### Customer Journey

- What factors are important when you implement a new technology? (e.g., cost, perceived value, long-term partnership, etc.)
    - How do you view sustainability?
- What risks do you see in the implementation of a new technology?
- What determines the value of a technology?

### Comparison of Different Payments

- What uncertainties do you see regarding the technology that the case company provides today?
- How does this uncertainty affect your willingness to pay? What determines how you prefer to pay for new technology? (e.g., capital investment, uncertainty, etc.)
- What type of payment model feels most reasonable for you?
    - How do you view paying an initial cost for ownership of the asset?
        - Why?
    - How do you view not owning the asset, and pay leasing that includes the right to use the technology?
        - Why?
    - How do you view the offering being delivered as a result (eg. only paying for the treatment)?
        - Why?
    - How would you respond to a solution where the offering is not the entire technical system, but only the rights to a patent?
        - Why?
- How do you view changes in the payment model after a certain period, when uncertainty about the technology has decreased?
- How flexible are you regarding changes in payment structure?
- What type of payment would you prefer to avoid, and why?

**If time allows**

- Are there any internal constraints that influence which model you can choose?
- Who typically makes the decision regarding the technology? Is it engineers, purchasers, or senior management?
- What information do you need, or what is most important for you, to feel confident in making a decision?

**Closing**

- Is there anything else you'd like to add that we haven't talked about?
- Ask if it would be okay if we contact them in case, we have any follow up questions regarding the discussion we have had today.
- Thank the respondent for taking time to participate in the study.